

Notes: Bolton and Harris (Econometrica, 1999)

Strategic Experimentation

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1 One-sentence summary

Bolton and Harris extend the classic two-armed bandit problem to many agents and show that information from experimentation is a public good, producing free-riding, but also an encouragement effect because current experimentation can accelerate future learning by others.

2 Big picture

2.1 Question

How do rational agents experiment when each agent can learn not only from her own risky action but also from the experiments of others?

2.2 Core object

The paper studies a continuous-time, many-agent, common-value two-armed bandit. Each player allocates time between a safe action with known payoff and a risky action with unknown common payoff.

2.3 Main friction

Information generated by risky experimentation is public: one player's experiment helps everyone update beliefs. This creates a free-rider problem because each agent would like others to bear the opportunity cost of experimentation.

2.4 Main countervailing force

There is also an encouragement effect. If an agent experiments today, she may bring forward the date at which future information from everyone becomes available, raising her continuation value.

2.5 Main equilibrium result

The unique symmetric Markov equilibrium has cutoff structure. Experimentation is zero below a lower cutoff, increases continuously over an intermediate belief region, and is full above an upper cutoff.

2.6 Welfare result

Equilibrium experimentation is below the team optimum. The free-rider effect dominates enough that even as the number of agents becomes large, the complete-information payoff is not attained.

2.7 Interpretation

The paper is a benchmark theory of social learning through costly experimentation. It shows why decentralized learning can be too slow even when all agents observe all experiments and all payoffs.

3 Incremental contribution of the paper

3.1 A multi-agent extension of the two-armed bandit

The paper turns the single-agent bandit into a strategic game in which many agents face the same unknown payoff and observe one another's experimentation. This creates an informational externality absent from the classic single-agent model.

3.2 Information as a public good

The central conceptual contribution is to treat experimental information as a public good. Agents benefit from others' risky actions, so private experimentation incentives do not equal social experimentation incentives.

3.3 Free-riding and encouragement in one model

The paper does not stop at the obvious free-rider channel. It also identifies the encouragement effect: more expected future experimentation by others raises the value of current experimentation because current learning can speed up access to the future information flow.

3.4 A tractable continuous-time equilibrium characterization

By formulating the problem in continuous time, the authors obtain Bellman equations and sharp characterizations of stationary Markov equilibria, including existence, uniqueness, cutoffs, and comparative statics.

3.5 Symmetric equilibrium despite public information

The paper shows that symmetric equilibrium requires time division or mixed strategies. In the indivisible pure-action version, symmetric equilibrium generally fails, but the mixed version is isomorphic to the time-division model.

3.6 A benchmark for social-learning models

Just as the single-agent two-armed bandit is a benchmark for individual active learning, this paper provides a benchmark for multi-agent active learning in research, exploration, technology adoption, consumer search, and investment discovery.

4 Conceptual framework

4.1 The economic environment

There are N identical, infinitely lived, risk-neutral players. At each instant, player i allocates a fraction $a_i \in [0, 1]$ of time to the risky action and $1 - a_i$ to the safe action.

The safe action has known payoff flow:

$$d\pi_i^0(t) = (1 - a_i)sdt + (1 - a_i)^{1/2}\sigma dZ_i^0(t).$$

The risky action has unknown underlying payoff $\mu \in \{l, h\}$:

$$d\pi_i^1(t) = a_i\mu dt + a_i^{1/2}\sigma dZ_i^1(t),$$

with $l < s < h$. Players observe all actions and all realized payoffs.

4.2 Beliefs as the state variable

The common belief is

$$p(t) = \Pr(\mu = h \mid \text{public history at } t).$$

The expected risky payoff is

$$m(p) = (1 - p)l + ph.$$

The break-even belief b solves $m(b) = s$.

4.3 Filtering equation

Define the information variance term

$$\Phi(p) = \left[p(1 - p) \frac{h - l}{\sigma} \right]^2.$$

Conditional on current information and experimentation choices, belief changes satisfy

$$E[dp(t)] = 0, \quad \text{Var}(dp(t)) = \left(\sum_{i=1}^N a_i \right) \Phi(p(t)) dt.$$

Detailed interpretation: Beliefs follow a martingale, but their variance rises with total experimentation. More experimentation means faster information arrival, not a predictable drift in beliefs.

4.4 Myopic payoff and full-information payoff

The myopic payoff is

$$u(p) = \max\{s, m(p)\}.$$

The full-information payoff is

$$\bar{u}(p) = (1 - p)s + ph.$$

The value of experimentation lies between these two payoffs.

5 Model setup and derivations

5.1 Individual Bellman equation

Given other players' strategies, player i solves:

$$u_i(p) = \max_{a_i \in [0,1]} \left\{ (1 - a_i)s + a_i m(p) + \frac{1}{r} \left(a_i + \sum_{j \neq i} a_j(p) \right) \Phi(p) \frac{u_i''(p)}{2} \right\}.$$

Detailed interpretation: The last term is the continuation value of information. It combines the discount factor $1/r$, the amount of total experimentation, the informativeness of an experiment, and the curvature of the value function in beliefs.

5.2 Opportunity cost versus shadow value of experimentation

Experimentation is privately optimal when

$$\frac{1}{r} \Phi(p) \frac{u_i''(p)}{2} > s - m(p).$$

The left-hand side is the shadow value of experimenting; the right-hand side is the current opportunity cost of using the risky arm rather than the safe arm.

5.3 Team problem

A planner maximizing average payoff chooses all agents' actions jointly:

$$u^*(p) = \max_{a \in [0,1]^N} \left\{ \frac{1}{N} \sum_{i=1}^N [(1 - a_i)s + a_i m(p)] + \frac{1}{r} \left(\sum_{i=1}^N a_i \right) \Phi(p) \frac{u^{*''}(p)}{2} \right\}.$$

The team solution has a single cutoff c^* : all agents play safe below c^* and all agents experiment above c^* .

Detailed interpretation: Because the planner internalizes the informational public good, the team experiments more aggressively than decentralized agents.

5.4 Symmetric equilibrium condition

In symmetric equilibrium, each player uses the same strategy $\alpha(p)$. Define the collective incentive to experiment:

$$\beta(p) = \frac{u(p) - s}{s - m(p)}.$$

The equilibrium experimentation rule is:

$$\alpha(p) = \begin{cases} 0, & \beta(p) < 0, \ p < b, \\ \beta(p)/(N-1), & 0 \leq \beta(p) \leq N-1, \ p < b, \\ 1, & \beta(p) > N-1 \text{ or } p \geq b. \end{cases}$$

Detailed interpretation: The intermediate region is the key strategic-experimentation region. Each player partially experiments because her private incentive is balanced against the free riding on others' experiments.

6 Main cases in the paper

6.1 Single-agent benchmark

With $N = 1$, the problem is the classic one-agent bandit. The agent experiments when beliefs are sufficiently optimistic and stops when the posterior falls below a cutoff.

6.2 Team-learning benchmark

For an N -player team, all informational benefits are internalized. The team cutoff c^* decreases with N because the opportunity cost of a unit of information is effectively shared across more agents.

6.3 Symmetric strategic equilibrium

For $N \geq 2$, there are two equilibrium cutoffs:

$$0 < c_t < c_T < b < 1.$$

Experimentation is:

$$\begin{aligned}\alpha(p) &= 0 && \text{for } p \leq c_t, \\ \alpha(p) &\in (0, 1) && \text{for } p \in (c_t, c_T), \\ \alpha(p) &= 1 && \text{for } p \geq c_T.\end{aligned}$$

6.4 Mixed-strategy interpretation

The divisible-time model is isomorphic to a mixed-strategy version of an indivisible-action model. The fraction of time spent experimenting can be reinterpreted as the probability of choosing the risky action.

6.5 Public randomization and taking turns

A public randomization device can correlate players' experiments, partially capturing the idea of taking turns. However, literal taking turns is not generally incentive compatible because a player assigned to the safe arm may want to deviate when experimentation is valuable.

7 Economic intuition

7.1 Why free-riding arises

Each agent bears the current cost of experimentation but shares the informational benefit with all agents. This makes the private marginal benefit of experimentation smaller than the social marginal benefit.

7.2 Why encouragement arises

If other agents will experiment in the future, then experimenting now can increase the chance that beliefs move into regions where those future experiments happen sooner. This raises the continuation value of one's own experimentation.

7.3 Why experimentation has cutoffs

When the belief p is too pessimistic, risky experimentation is too costly. When p is optimistic, the risky arm is worth using both for payoff and information. In between, agents partially experiment because information is valuable but current payoffs favor the safe arm.

7.4 Why complete information is not reached even with many agents

As N rises, total potential experimentation rises, but individual incentives to free ride also rise. The paper shows that even as $N \rightarrow \infty$, the complete-information payoff is not attained for interior beliefs.

8 Identification and theoretical design

8.1 What the paper identifies theoretically

The model isolates a pure public-good externality in learning. There are no strategic complementarities in payoffs, no private values, no asymmetric costs, and no hidden actions after the fact. The only strategic interaction is through the information generated by experimentation.

8.2 Why the common-value assumption matters

Because the risky payoff is common across players, any experimental signal is valuable to all. This is what creates the informational public good. If values were purely private, experimentation would be less socially shared.

8.3 Why continuous time matters

Continuous time converts the experimentation problem into a diffusion filtering problem with tractable Bellman equations. This allows the authors to characterize cutoffs, uniqueness, and comparative statics more cleanly than in a discrete-time many-agent bandit.

8.4 Observable implications

The theory predicts:

- agents experiment less than a team would;
- experimentation rises with the belief that the risky arm is good;
- experimentation can be partial in intermediate belief states;
- value functions improve with more agents even though total experimentation need not be monotone;
- public coordination can change correlation in experiments but not remove incentive constraints.

8.5 Important limitations

The baseline model has two states, one safe arm, one risky arm, common values, perfect public observation, and symmetric players. These restrictions make the benchmark clean but abstract from many real environments where signals are private, payoffs are heterogeneous, or experimentation choices produce different types of information.

9 Tables and figures

9.1 Notes-created Table 1: Core model objects; Notes-created Table 2: Main mechanisms

Object	Meaning
p	posterior probability risky arm is high
$m(p)$	expected risky payoff
b	belief where $m(p) = s$
$\Phi(p)$	informativeness of risky experimentation
a_i	player i 's experimentation intensity
$u(p)$	equilibrium value function
c_t, c_T	lower and upper equilibrium cutoffs

Notes-created Table 1: Core model objects.

Mechanism	Role
Free-riding	lowers private experimentation below team level
Encouragement	raises value of current experimentation by accelerating future information
Cutoffs	partition safe, partial, and full experimentation regions
Mixed strategies	reinterpret time shares as probabilities
Public randomization	coordinates actions but does not remove incentive constraints

Notes-created Table 2: Main mechanisms.

Read: Table 1 defines the key objects needed to read the model. Table 2 summarizes the economic forces driving equilibrium.

Economic message: The model is a theory of information provision. The private action is an experiment, but the information generated by that experiment is shared by everyone.

Detailed interpretation: The two most important concepts are $\Phi(p)$ and the cutoffs. $\Phi(p)$ tells us when information is potentially valuable: it is highest when the state is uncertain and zero when the state is already known. The cutoffs then show how private incentives translate this information value into action.

9.2 Notes-created Table 3: Main formal results; Notes-created Table 4: Comparative statics

Result	Content
Lemma 1	belief is a martingale; variance rises with total experimentation
Lemma 5	team problem has a single cutoff
Theorem 9	symmetric equilibrium characterized by collective incentive
Lemma 10	equilibrium value lies below team value
Theorem 11	unique symmetric equilibrium exists
Theorem 12	payoffs rise with patience and number of players

Notes-created Table 3: Main formal results.

Comparative static	Interpretation
Higher patience	raises experimentation value and payoff
More players	raises payoff via encouragement effect
$N \rightarrow \infty$	complete-information payoff still not reached
Higher signal precision	raises informational value of experimentation
Pessimistic p	safe action dominates
Optimistic p	risky action dominates

Notes-created Table 4: Comparative statics.

Read: These two notes-created tables summarize the theorem-level structure of the paper.

Economic message: The formal results make one point repeatedly: more learning opportunities help, but public-good incentives prevent decentralized experimentation from becoming fully efficient.

Detailed interpretation: Theorem 12 is especially important. It would be tempting to think that many agents eliminate the learning problem, because someone will experiment. The result says no: many agents also magnify free-riding incentives, so complete-information payoffs are not achieved.

9.3 Notes-created Table 5: Original visuals status

Original visual type	Status in Bolton and Harris (1999)
Empirical figures	none
Empirical tables	none
Theoretical diagrams	none
Notes-created tables	used to summarize objects, mechanisms, and propositions

Notes-created Table 5: Original visuals status. The original paper is a theory paper with no original tables or figures, so the notes use compact summary tables instead.

10 Short summary

- The paper studies many agents who can all learn from each other's experiments.
- Information generated by risky experimentation is a public good.
- The decentralized equilibrium features free-riding: agents experiment less than the team optimum.
- There is also an encouragement effect: future experimentation by others can raise current experimentation incentives.
- The team problem has a single cutoff; symmetric equilibrium has lower and upper cutoffs with partial experimentation in between.
- Symmetric equilibrium is unique.
- Equilibrium payoffs rise with patience and with the number of players.
- Complete-information payoffs are not attained even as the number of players becomes large.
- The divisible-time model is equivalent to a mixed-strategy indivisible-time model.
- Public randomization can coordinate actions but cannot fully implement taking turns.

11 Conclusion

11.1 Core conclusion

Bolton and Harris show that decentralized experimentation is inefficient when information is publicly observed. Individual agents underprovide experimentation because they can learn from others without paying the full opportunity cost of experimentation themselves.

11.2 Why the result is not just free-riding

The paper’s subtlety is that strategic experimentation also generates encouragement. If others will experiment in the future, then experimenting today may speed up the arrival of future information. This effect mitigates free-riding and makes payoffs increase with the number of agents.

11.3 Incremental contribution revisited

The paper builds the canonical strategic version of the two-armed bandit. It shows how the central insights of single-agent active learning change when experimental information is publicly shared across many agents.

11.4 Implications for economics

The model applies to R&D races, technology adoption, mineral exploration, consumer search, investment discovery, and macroeconomic information acquisition. In each setting, private incentives to gather information may be too weak because the information spills over to competitors or peers.

11.5 Limitations and future work

The baseline has only two states, one safe action, one risky action, identical players, and perfectly public observation. The authors note that richer models can allow more states, different signals and payoffs, and multiple risky actions with different information structures.

11.6 Takeaway

Strategic experimentation is a public-good problem. More agents create more potential information, but also more incentives to free ride. The equilibrium balances free-riding and encouragement, producing experimentation that is socially useful but still inefficiently slow.